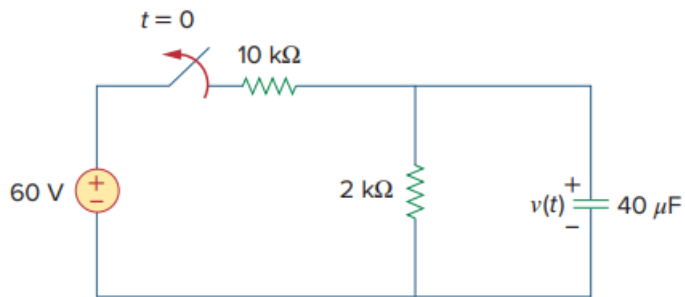


Problem

The switch in Fig. 7.86 has been closed for a long time, and it opens at $t = 0$. Find $v(t)$ for $t \geq 0$.

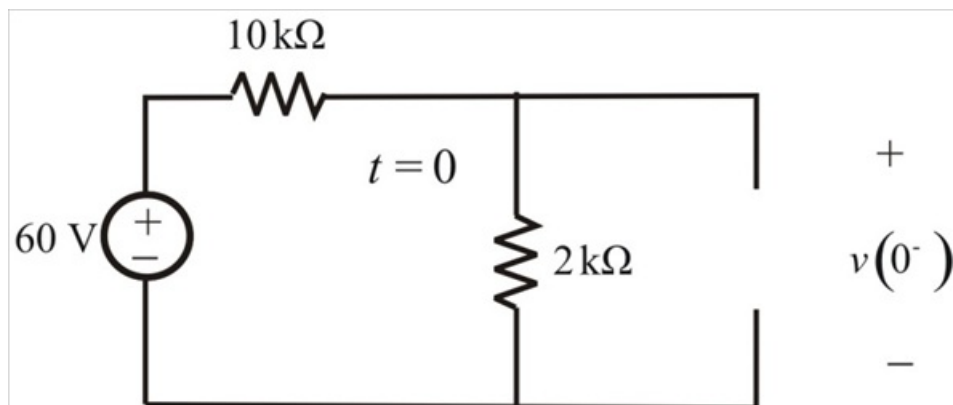
Fig. 7.86:



Step-by-step solution

Step 1 of 3

Switch is opens at $t = 0$ before that it was closed for a long time. Capacitor acts like open circuit to DC. Hence the voltage across the capacitor just before $t = 0$ is obtained be



Step 2 of 3

By voltage division rule

$$v(0^-) = 60 \times \frac{2k}{10k + 2k}$$

$$v(0^-) = 10 \text{ V}$$

Capacitor voltage can't change instantaneously. So

$$v(0^-) = v(0) = 10 \text{ V}$$

Switch is opens at $t = 0$. For $t > 0$ circuit is source free network. So capacitor voltage is

$$v(t) = v(0)e^{-\frac{t}{\tau}}$$

$$v(t) = 10e^{-\frac{t}{\tau}}$$

Where

$$\tau = R_{eq}C$$

Calculate the value of equivalent resistance seen from the capacitance at $t > 0$.

$$R_{eq} = 2 \text{ k}\Omega$$

Step 3 of 3

Calculate the time constant.

$$\tau = R_{eq}C$$

Substitute 2×10^3 for R_{eq} , and 40×10^{-6} for C.

$$\tau = 2 \times 10^3 \times 40 \times 10^{-6}$$

$$= 80 \times 10^{-3}$$

$$= 0.08 \text{ sec}$$

Therefore,

$$v(t) = 10e^{-\frac{100}{8}t} \text{ V}$$

$$= 10e^{-12.5t} \text{ V}$$

Therefore, the value of voltage $v(t)$ for $t > 0$ is $\boxed{10e^{-12.5t} \text{ V}}$.